Light

GCSE Computer Science - Fetch-Decode-Execute Cycle and Embedded Systems (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

How often does the fetch-decode-execute cycle repeat during normal computer operation?

- ○ **A.** Continuously, billions of times per second while the computer is running
- ○ **B.** Once when the computer first starts up
- ○ **C.** Only when the user presses a key
- ○ **D.** Once per minute to check for updates

Q2 (1 mark)

Which CPU component is primarily responsible for the decode stage of the FDE cycle?

- ○ **A.** The arithmetic logic unit
- ○ **B.** The hard disk drive
- ○ **C.** The graphics card
- ○ **D.** The control unit

Q3 (1 mark)

Which of the following could happen during the execute stage?

- ○ **A.** The instruction is copied from RAM to the CPU
- ○ **B.** A calculation is performed by the ALU or data is read from memory
- ○ **C.** The binary instruction is interpreted
- ○ **D.** The computer is switched off

Q4 (1 mark)

Which operation would the ALU perform when a program checks if a user's password is correct?

- ○ **A.** Division
- ○ **B.** Multiplication
- ○ **C.** A logical comparison to check if two values are equal
- ○ **D.** Storing the password permanently

Q5 (1 mark)

How does the control unit communicate with other CPU components?

- ○ **A.** By sending electrical control signals through the control bus
- ○ **B.** By displaying messages on the screen
- ○ **C.** By making audible sounds
- ○ **D.** By sending emails to the components

Q6 (1 mark)

What unit is used to measure CPU clock speed?

- ○ **A.** Bytes
- ○ **B.** Metres per second
- ○ **C.** Watts
- ○ **D.** Hertz (Hz), typically gigahertz (GHz)

Q7 (1 mark)

What is the specific purpose of the address bus?

- ○ **A.** To carry the actual data being transferred
- ○ **B.** To carry memory location addresses from the CPU to memory
- ○ **C.** To provide power to the CPU

- ○ **D.** To cool the processor

Q8 (1 mark)

What is a processor core?

- ○ **A.** An independent processing unit within the CPU that can execute its own instructions
- ○ **B.** The metal casing around the CPU
- ○ **C.** The cooling fan attached to the processor
- ○ **D.** The socket where the CPU connects to the motherboard

Q9 (1 mark)

What is a key difference between an embedded system and a desktop computer?

- ○ **A.** Desktop computers are always more expensive
- ○ **B.** Desktop computers cannot run software
- ○ **C.** Embedded systems are always connected to the internet
- ○ **D.** Embedded systems are designed for specific tasks while desktop computers are general-purpose

Q10 (1 mark)

Which device contains an embedded system?

- ○ **A.** A digital camera
- ○ **B.** A standard desktop PC
- ○ **C.** A laptop computer
- ○ **D.** A computer server

Q11 (1 mark)

Why is cache memory positioned close to the CPU?

- ○ **A.** To make the computer look neater
- ○ **B.** To reduce the distance electrical signals must travel, enabling faster data access
- ○ **C.** To increase storage capacity
- ○ **D.** To connect to the internet more quickly

Q12 (1 mark)

Why do registers have very small storage capacity compared to RAM?

- ○ **A.** Because registers are made from cheaper materials
- ○ **B.** Because registers can only store numbers
- ○ **C.** Because the computer manufacturer wants to save money
- ○ **D.** Because registers only need to hold the data immediately being processed, and making them larger would slow them down

Q13 (2 marks)

During the fetch stage, how does the CPU know which instruction to fetch next?

- ○ **A.** The user types in which instruction to run
- ○ **B.** Instructions are always fetched in random order
- ○ **C.** A register called the program counter holds the memory address of the next instruction to fetch
- ○ **D.** The hard drive sends instructions automatically

Q14 (2 marks)

A student writes a program to calculate the average of three test scores. Which CPU component performs the actual calculations?

- ○ **A.** The ALU, which adds the scores together and divides by three
- ○ **B.** The control unit
- ○ **C.** The system clock
- ○ **D.** The data bus

Q15 (2 marks)

Why can manufacturers not simply increase clock speed indefinitely to make CPUs faster?

- ○ **A.** There are no limitations on clock speed
- ○ **B.** Higher clock speeds are illegal in most countries

- ○ **C.** Clock speed has no effect on performance
- ○ **D.** Higher clock speeds generate more heat and consume more power, creating cooling and energy constraints

Q16 (2 marks)

Why do many embedded systems have minimal or no traditional user interface?

- ○ **A.** Because embedded systems cannot process any input
- ○ **B.** Because they perform dedicated tasks automatically or with simple controls, not requiring keyboard and screen interfaces
- ○ **C.** Because user interfaces are illegal for embedded systems
- ○ **D.** Because embedded systems are always hidden underground

Q17 (2 marks)

A computer has a 64-bit data bus. What does this mean for data transfer?

- ○ **A.** The computer can only store 64 bits of data total
- ○ **B.** Data can only be transferred once every 64 seconds
- ○ **C.** 64 bits of data can be transferred in a single bus operation, enabling faster data movement than narrower buses
- ○ **D.** The bus is 64 metres long

Q18 (2 marks)

A video game is designed to run on a single thread. How does this affect the benefit of having multiple CPU cores?

- ○ **A.** More cores will make the game run proportionally faster
- ○ **B.** The game can only use one core at a time, so additional cores provide little benefit for that specific game
- ○ **C.** The game will not run at all on multi-core processors
- ○ **D.** Each core will run a separate copy of the game

Q19 (2 marks)

Why is real-time response particularly important for embedded systems in vehicles?

- ○ **A.** Delays in processing could cause safety-critical systems like brakes or airbags to respond too slowly, risking lives
- ○ **B.** Real-time response is not important for vehicle systems
- ○ **C.** Real-time response only matters for entertainment systems
- ○ **D.** Vehicle embedded systems do not need to respond to any inputs

Q20 (2 marks)

What is a 'cache hit' and why is it beneficial?

- ○ **A.** A cache hit damages the CPU and should be avoided
- ○ **B.** A cache hit only occurs when the computer crashes
- ○ **C.** A cache hit means the cache is full and needs to be emptied

○ **D.** A cache hit occurs when requested data is found in cache, providing faster access than retrieving from main memory

Q21 (2 marks)

How does the control unit use the control bus during CPU operations?

○ **A.** The control bus carries data values between components

○ **B.** The control unit sends timing and operation signals through the control bus to coordinate component actions

○ **C.** The control bus stores programs permanently

○ **D.** The control bus only operates when the computer is switched off

Q22 (2 marks)

Why do CPUs use registers instead of accessing RAM directly for all data?

○ **A.** Registers are much faster than RAM, so using them for active data dramatically reduces processing delays

○ **B.** RAM cannot store numerical data

○ **C.** Registers are less expensive than RAM

○ **D.** RAM is located outside the computer

Q23 (2 marks)

Why do many embedded systems need to operate on very low power?

- ○ **A.** High power usage is illegal for embedded systems
- ○ **B.** Power consumption has no effect on embedded system design
- ○ **C.** Embedded systems cannot connect to any power source
- ○ **D.** Many embedded systems run on batteries or in environments where power efficiency is critical for extended operation

Q24 (3 marks)

A program contains the instruction to subtract the value in memory location 200 from the value in memory location 100 and store the result in location 300. Trace through how the FDE cycle processes this instruction.

- ○ **A.** The subtraction happens automatically without any CPU involvement
- ○ **B.** Fetch: retrieve the subtract instruction from memory; Decode: interpret the instruction to identify locations 100, 200, and 300 and the subtract operation; Execute: read values from locations 100 and 200, perform subtraction in ALU, write result to location 300
- ○ **C.** Only the execute stage is needed; fetch and decode are skipped for mathematical operations
- ○ **D.** The values are stored directly in the ALU permanently

Q25 (3 marks)

Analyse why computers are designed with both cache and RAM rather than using only one or the other.

- ○ **A.** There is no benefit to having both; engineers include both purely for tradition

○ **B.** Cache alone would provide sufficient storage for all programs and data

○ **C.** Cache provides speed for frequently-accessed data but is too expensive and small for all storage needs; RAM provides larger capacity at lower cost but is slower. The combination balances speed, capacity, and cost effectively

○ **D.** RAM alone is fast enough for all purposes; cache serves no useful function

Q26 (3 marks)

A medical device company is designing a pacemaker. Explain why an embedded system is essential rather than using a general-purpose computer.

○ **A.** General-purpose computers would be equally suitable for pacemakers

○ **B.** Embedded systems are only used in devices that cost less than 10 pounds

○ **C.** Pacemakers do not contain any computer systems

○ **D.** An embedded system provides the reliability, small size, low power consumption, instant response, and dedicated function essential for a life-critical device implanted in the human body

Q27 (3 marks)

Analyse how the control unit, ALU, registers, and buses collaborate to execute a simple addition instruction.

○ **A.** Each component works independently with no coordination

○ **B.** Only the ALU is involved; other components are not used for

calculations

○ **C.** The control unit directs the operation, sending signals via the control bus; data moves through the data bus to registers; the ALU performs the addition using register contents; results are stored in a register, with the control unit coordinating timing throughout

○ **D.** The buses perform the calculation while other components observe

Q28 (3 marks)

A software developer is optimising a video encoding program. Explain how they could modify their code to better utilise a multi-core processor and what benefits this would provide.

○ **A.** The developer should divide the video into separate segments that can be encoded simultaneously by different cores; this parallel processing reduces total encoding time roughly proportional to the number of cores available

○ **B.** Software cannot be modified to use multiple cores

○ **C.** Using more cores would slow down the encoding process

○ **D.** Multi-core processors can only run one program at a time

Q29 (3 marks)

Explain the relationship between main memory and the CPU during the fetch-decode-execute cycle, including what is fetched and why main memory is essential.

○ **A.** Main memory is not involved in the FDE cycle

○ **B.** Main memory stores only temporary calculations during execution

○ **C.** Main memory holds the program instructions and data that the CPU fetches during the fetch stage; without main memory, the CPU would have no instructions to execute or data to process, making the FDE cycle impossible

○ **D.** The CPU contains all necessary programs internally and never accesses main memory

Q30 (3 marks)

A smart home company is deciding whether to use embedded systems or general-purpose computers for: (1) smart light switches, (2) a home automation hub, (3) motion sensors. Evaluate the appropriate choice for each.

○ **A.** General-purpose computers should be used for all three devices

○ **B.** Embedded systems should be used for all three devices regardless of function

○ **C.** Smart light switches and motion sensors suit embedded systems due to simple dedicated functions and power constraints; the home automation hub might benefit from a more capable embedded system or small general-purpose computer since it must run varied applications and integrate multiple devices

○ **D.** The choice between embedded and general-purpose systems has no practical impact

Total: **55** marks • Time: **30 min**

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